5C34 : ADVANCED AI
Fréchet Audio Distance: Evaluating Generative Music Models

Alba Carrilo Martinez
Haoran Zhang
Ishan Samir Thoke

Trinity College Dublin
MSc. Electronic Information Engineering
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Trinity College Dublin
Coláiste na Tríonóide, Baile Átha Cliath
The University of Dublin

Slide 1: The Evaluation Problem

We can hear bad music. But can machines measure it?

Generative music models (MusicGen, MusicLM, Noise2Music) produce audio — how do you objectively evaluate quality?

Can't use human listeners at scale as it is subjective, expensive, slow.

Need a metric that:

- requires no reference track per-sample,
- correlates with human perception,
- works across models.

Three naive approaches that fail:

- SDR (Signal-to-Distortion Ratio): needs a ground-truth reference per clip.
- L2 distance in waveform space: penalises time-shifts, not perceptual distortion.
- Cosine similarity of raw embeddings: reported to correlate poorly with human judgement.

Slide 2: Inspiration — FID in Image Generation

Before FAD, there was FID (Fréchet Inception Distance).

The same problem was solved for images in 2017:

- Extract deep features from real and generated images using Inception-v3.
- Model both distributions as Gaussians.
- Measure Wasserstein-2 distance between them.

FAD is the audio equivalent, proposed by Kilgour et al. (Google, 2019).

Remember GANs and diffusion? FID is the standard metric for both. FAD carries the same philosophy into the audio domain.

Slide 3: How FAD Works

Four steps:

1. Pass reference audio (real music) through VGGish.
2. Pass generated audio through the same VGGish.
3. Fit a multivariate Gaussian (μ, Σ) to each embedding set.
4. Compute the Fréchet distance between the two distributions.

$$\mathrm{FAD} = \|\mu_r - \mu_g\|^2 + \mathrm{tr}\left(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2}\right)$$

Lower FAD = generated audio statistically closer to real music.

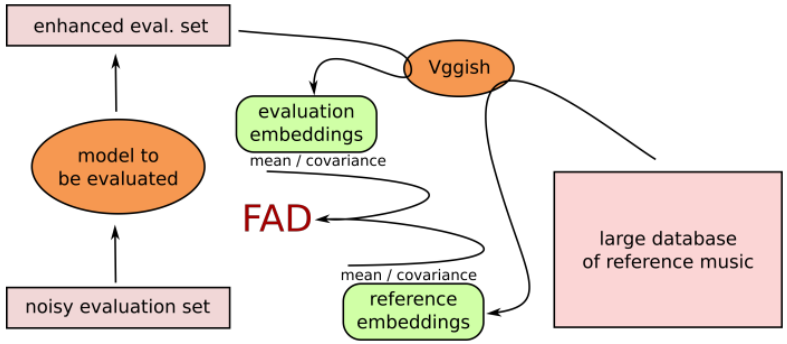

Figure 1. FAD computation pipeline.

Slide 4: Why VGGish?

VGGish is a VGG-style CNN trained on YouTube-8M.

- Input: 96 frames $\times$ 64 log-mel features (1 second of audio).
- Output: 128-dimensional embedding from the penultimate layer.
- Trained on around 3,000 audio classes, so it learns perceptually meaningful features.

The key insight:
By embedding audio into a perceptual feature space before measuring distance, FAD captures *how audio sounds* rather than *how audio looks as a waveform*.

Correlation with human perception:
Reported correlation is around $r = 0.52$, significantly outperforming SDR ($r = 0.39$) and signal-level metrics.

Slide 5: FAD in Practice — Music Generation

FAD became the standard evaluation metric for major music generation models.

Model	**FAD Score**
Ground truth reference	0.0
MusicGen (large)	$\sim$1.8
AudioLDM	$\sim$4.0
Random noise	$\gg 20$

How MusicGen used FAD:
Meta evaluated MusicGen by computing FAD against the MusicCaps dataset as the reference set, and this is the same pipeline we'd use for Catharsis.

Slide 6: Problem 1 — Sample Size Bias

FAD scores are not stable with small sample sizes.

- More generated samples → lower FAD score, regardless of quality.
- Different models have different bias rates, making comparisons unfair.
- Many papers do not clearly report sample sizes, so results become harder to reproduce.

Fix: FAD$_\infty$

- Sample at multiple sizes N.
- Fit a linear regression.
- Extrapolate to $N \to \infty$ to get an unbiased estimate.

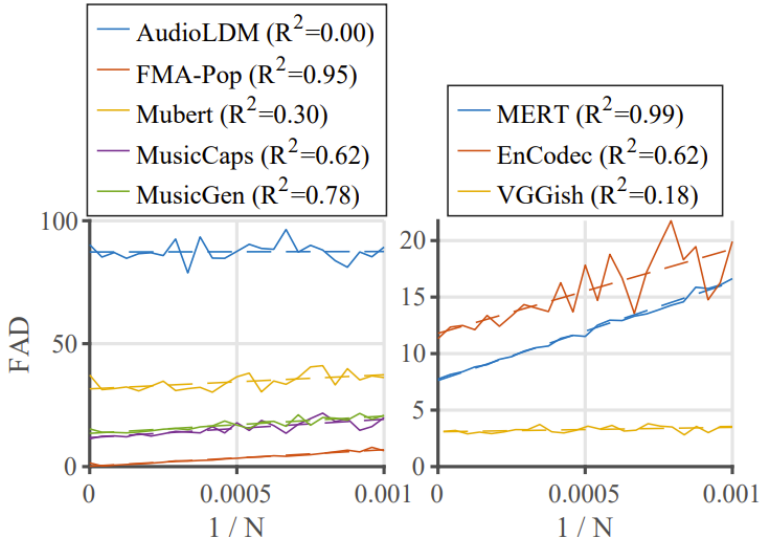

Fig2. FAD decreases as sample size increases.

Slide 7: Problem 2 — Embedding Choice Matters

VGGish was trained on general YouTube audio and not specifically on music.

- CLAP and LAION-CLAP outperform VGGish for predicting MusicCaps quality labels.
- Embedding choice strongly affects the final FAD result.
- Joint audio-text embeddings capture both acoustic quality and musical characteristics.

Emb.	Trained On	Corr.
VGGish	General audio	Low
MERT	Music only	Medium
CLAP	Audio-text	High
LAION-CLAP	Music-text	Highest

Figure 3. Comparison of embeddings for generative music evaluation.

Slide 8: Problem 3 — FAD Doesn't Measure Relevance

FAD only measures audio quality — not whether the music matches the prompt.

A model could generate perfect-sounding jazz to a "sad piano" prompt and still score very well on FAD.

You need a separate metric for text-audio alignment.

Two complementary metrics in practice:

- FAD $\rightarrow$ Does it sound like real music?
- CLAP Score $\rightarrow$ Does it match the text description?
- KLD (KL Divergence on classifier outputs) $\rightarrow$ Does it have the right audio characteristics?

For Catharsis specifically: FAD measures whether the output sounds musical; you'd also need something to verify whether the intended emotion was captured correctly.

Slide 9: The Challenger — KAD (2025)

"No More FAD!" — Kernel Audio Distance was proposed in 2025.

The problem with FAD's Gaussian assumption:

- Real audio embeddings are not Gaussian; in practice they are often multimodal.
- Forcing a Gaussian fit loses distributional nuance.

KAD's fix:

- Replace the Gaussian assumption with Maximum Mean Discrepancy (MMD) using a kernel.
- Non-parametric, so no distribution assumption is needed.
- More robust to small samples and more sensitive to subtle quality differences.

Still early: FAD remains the standard in most papers and toolkits, but KAD is gaining attention.

Slide 10: Summary & Open Problems

FAD's contribution:

- Reference-free, distribution-level metric: the right abstraction for generative models.
- Direct audio analogue of FID from your course's generative models week.

What's still unsolved:

- No single metric captures both quality *and* relevance.
- The Gaussian assumption is theoretically questionable.
- Standardisation remains open: which embedding and which reference set?
- There is still no accepted metric for emotional accuracy in music generation.

Thank you!

Alba Carrilo Martinez
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